PROBLEM DA4

Terminal Velocity of Slime Particles

Original statement · DA4.pdf

Π theorem | Change of fluid | Terminal velocity

OFFICIAL PROBLEM STATEMENT

Chemical and hydraulic engineers are often interested in computing the settling velocity of a suspension of particulates in a given liquid. For dilute suspensions, a variable that is directly related with this settling velocity is the so-called terminal velocity, u_t, which is the constant falling velocity of an isolated particle moving in an infinite fluid under the action of gravity. For the particular case of slime sedimentation in a river, the slime particles are spheres of average diameter $D_s = 30$ μm and average mass $m_s = 0.2$ μg, much too small to be handled in the lab. To guide the design of an experiment to determine u_t, the following steps are suggested:

1. Use the Π theorem to reduce the parametric dependence of the terminal velocity.

2. To enable the use of larger particles in the experiment, explore the replacement of water with glicerine ($\nu_g/\nu_w = 680$, $\rho_g/\rho_w = 1.25$) and find the corresponding values of the diameter D_e and mass m_e for the resulting experimental particle.

3. For this particle, the experimental measurements give $u_e = 10^{-3}$ m/s. Obtain the terminal velocity for an average slime particle.

Source: DA4.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; physical similarity, Eqs. 6.57-6.59.

Worked solution

Hypotheses, Data and Dimensional Table

- At terminal velocity the particle does not accelerate: its weight, the buoyancy and the drag balance. The weight enters through mg, the buoyancy through ρg times the volume ($\propto D^3$), the drag through ρ, μ, D and u_t.
- The particle is a sphere in an unbounded liquid: no free surface, no walls.

Parameter	Meaning	Dimensions
u_t	terminal velocity (unknown)	LT^{-1}

Parameter	Meaning	Dimensions
D	particle diameter	L
m	particle mass	M
g	gravity	LT^{-2}
ρ	liquid density	ML^{-3}
μ	liquid viscosity	$ML^{-1}T^{-1}$

Application of the Π Theorem

$$u_t = u_t(D, m, g, \rho, \mu)$$

$n = 5$. The key takes (D, g, ρ): rows $D(1,0,0)$, $g(1,0,-2)$, $\rho(-3,1,0)$ in (L, M, T), with $\det = 1 \cdot (0 \cdot 0 - (-2) \cdot 1) - 0 + 0 = 2 \neq 0$, so $k = 3$ and two groups remain.

Velocity. $[u_t] = [D]^a[g]^b$: $T:\ -1 = -2b$, so $b = 1/2$; $L:\ 1 = a + b$, so $a = 1/2$: the scale $\sqrt{gD}$.

Mass. $[m] = [\rho][D]^3$: the group $m/(\rho D^3)$, the ratio of the particle density to the liquid density up to the factor $\pi/6$.

Viscosity. $[\mu] = [\rho]^a[g]^b[D]^c$: $M:\ 1 = a$; $T:\ -1 = -2b$, so $b = 1/2$; $L:\ -1 = -3a + b + c = -3 + \frac{1}{2} + c$, so $c = 3/2$: the group $\mu/(\rho g^{1/2} D^{3/2}) = \nu/(g^{1/2} D^{3/2})$.

$$\frac{u_t}{\sqrt{gD}} = f\left(\frac{m}{\rho D^3}, \frac{\mu}{\rho g^{1/2} D^{3/2}}\right)$$

Physical Similarity in Glycerine

Subscript s for the slime particle in water (ρ_w, ν_w), e for the experiment in glycerine (ρ_g, ν_g); same g. Both groups must be equal.

Viscosity group:

$$\frac{\nu_w}{g^{1/2} D_s^{3/2}} = \frac{\nu_g}{g^{1/2} D_e^{3/2}} \implies \left(\frac{D_e}{D_s}\right)^{3/2} = \frac{\nu_g}{\nu_w} \implies D_e = \left(\frac{\nu_g}{\nu_w}\right)^{2/3} D_s$$

$680^{2/3} = \exp\left(\tfrac{2}{3}\ln 680\right) = \exp(4.348) = 77.33$, so

$$D_e = 77.33 \times 30\ \mu\text{m} = 2320\ \mu\text{m} = 2.32\ \text{mm}$$

Mass group:

$$\frac{m_s}{\rho_w D_s^3} = \frac{m_e}{\rho_g D_e^3} \implies m_e = \frac{\rho_g}{\rho_w}\left(\frac{D_e}{D_s}\right)^3 m_s = 1.25 \times 77.33^3 \times 0.2\ \mu\text{g}$$

$77.33^3 = 4.624 \times 10^5$, so $m_e = 1.25 \times 4.624 \times 10^5 \times 0.2\ \mu\text{g} = 1.156 \times 10^5\ \mu\text{g} = 0.116$ g. The experimental particle is a sphere of 2.3 mm and about 0.12 g, easy to handle.

Terminal Velocity of the Slime

With both groups equal, the dimensionless velocity is equal:

$$\frac{u_t}{\sqrt{g D_s}} = \frac{u_e}{\sqrt{g D_e}} \implies u_t = \left(\frac{D_s}{D_e}\right)^{1/2} u_e = \left(\frac{1}{77.33}\right)^{1/2} \times 10^{-3}\ \text{m/s}$$

$$u_t = 0.1137 \times 10^{-3}\ \mathrm{m/s} = 0.1137\ \mathrm{mm/s}$$

Answers and Checks

- Reduced law: $u_t/\sqrt{gD} = f\left(m/(\rho D^3),\ \mu/(\rho g^{1/2} D^{3/2})\right)$.
- Experimental particle: $D_e = 2.32$ mm, $m_e = 0.116$ g.
- Slime terminal velocity: $u_t = 0.114$ mm/s.
- Dimensions: $\nu/(g^{1/2}D^{3/2})$ is $L^2T^{-1}/(L^{1/2}T^{-1} \cdot L^{3/2}) = 1$.
- Consistency: equal $m/(\rho D^3)$ means the particle-to-liquid density ratio is the same in both systems, so the net weight per unit volume scales with the liquid density, as it must for the same balance of weight, buoyancy and drag.
- Check against the official key: DA4.pdf gives $D_e = 2.32$ mm and $u_t = 0.1137$ mm/s; the key writes $m_e = (\rho_g/\rho_w)(D_e/D_s)^3 m_s$ without the number, which is 0.116 g.